# Dynamic Data Acquisition of Stool Radius

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***Abstract*—This experiment explored digital data acquisition, resolution limits, and uncertainty by measuring the radius of the footrest of a lab stool using three methods. These methods included using a Wheatstone bridge voltage, an amplified voltage, and a manual measuring tape. For the first two methods, a SADI 2.7 DAQ recorded data from a strain gauge through a LABview program which required careful selection of ADC sample windows to optimize resolution and avoid saturation. The third method measured the circumference of the footrest with a tape measure and used a hand calculation to find the radius. The methods using the Wheatstone bridge voltage and amplified voltage yielded radius results of 251.79 ± 39.00 millimeters and 250.06 ± 22.58 millimeters, respectively, while the tape measure method recorded a radius of 254.33 ± 0.0796 millimeters. Despite the similar accuracy of these results, the strain gauge methods were much less precise due to a larger stack up of uncertainties from multiple sources, sensor noise, and potential other issues.**

***Index Terms*— Amplifier, Data Acquisition Device, Strain Gauge, Wheatstone Bridge**

## I. Introduction

THE main objective of this lab was to explore digital data acquisition, resolution limits, and uncertainty by measuring the radius of the footrest of a lab stool with various methods and comparing the results. To achieve this objective, a SADI 2.7 DAQ was constructed and connected to a Wheatstone bridge (WSB) and amplifier to record data from a strain gauge affixed to a feeler gauge through a LabVIEW program. A basic knowledge of the functionality of these devices was necessary to understand and carry out the experiment.

A transducer is a device that changes one physical phenomena into another [3]. For this lab, a strain gauge attached to a feeler gauge acted as a transducer to convert a voltage reading to a strain estimation. When the strain gauge was flexed, the WSB read the resistance change and expressed it as a voltage drop. This voltage could be scaled up by the amplifier and adjusted with a trim potentiometer to a desired value.

The analog voltage generated by the WSB and strain gauge was measured by a Data Acquisition device (DAQ). The DAQ has an Analog to Digital Converter (ADC) that enables it to measure the voltage by taking an analog signal, approximating it with a possible value, and returning a discrete representation both in time and magnitude [3]. However, there are limitations to this process. If the analog value is out of the range of the limited number of values the ADC can approximate, then the ADC is considered saturated [3]. If this occurs, then input signals can be amplified to avoid saturation and increase resolution [3].

The resolution for an ADC is defined by the sample windows of the DAQ that allow measurements of voltages over different ranges. The sample windows used in this lab are ±0.64V, ±1.28V, ±2.56V, ±5.12V, and ±10.24V. Each one of these sample windows has a different resolution that can be calculated by the equation below (1), where $V_{max}$ is the maximum voltage of the ADC, $V_{min}$ is the minimum voltage of the ADC, and $n$ is the number of bits the ADC has [3].

$$Resolution = \frac{(V_{max} - V_{min})}{2^n - 1} \tag{1}$$

The resolution defines the smallest incremental voltage that can be recognized by the ADC with a higher resolution corresponding to a lower range of data values it can gather. The SADI 2.7 DAQ used in this lab has an $n$ value equal to 14 bits which means it can return 16,383 unique values [3]. Choosing the correct sample window for the voltages being measured is critical to receive accurate data and avoid saturation.

A strain gage is used to measure strain in a feeler gage when bent around a circular object [1]. In the case of this lab, a strain gage glued to a feeler gage was taped around the curvature of the footrest of a stool to collect data to approximate the radius. First, the strain is measured using the equation below (2) where $\rho$ is the curvature of the neutral axis and $y$ is the distance from the neutral axis of the feeler gauge and a point of interest [5].

$$\varepsilon = \frac{y}{\rho} \tag{2}$$

The variable $y$ in this equation depends on the thickness of the feeler gauge, $t$, and the distance from the surface of the feeler gauge to the center of the strain gauge, $d$, as expressed in the equation below (3).

$$y = d + \frac{t}{2} \tag{3}$$

Now that the strain can be measured with the above two equations, a third equation relates the previously defined variables to calculate the inside radius of the stool, $ri$, shown below (4).

$$ri = \frac{d + t/2}{\varepsilon} - \frac{t}{2} \quad (4)$$

After the devices operating with the LABview program made the calculations above, a tape measure was then used to manually measure the circumference, *C*, of the stool and thus hand calculate the inside radius, *ri*, with the equation below (5).

$$ri = \frac{C}{2\pi} \quad (5)$$

Uncertainty serves as an estimation of the range of error in a value [4]. Sources of uncertainty in this lab included variability in material properties and variability in the sensors and measurement systems used [4]. The uncertainties for the methods used in this experiment were calculated and compared to provide insight into the accuracy and precision of the results.

## II. Procedure

The required materials for this lab were a DAQ, Wheatstone bridge, strain gage affixed to a feeler gage, jumper wires, a computer capable of running LABview, a USBC chord, a measuring tape, and a stool. Before beginning the experiment, the DAQ was properly connected to the Wheatstone bridge and amplifier with jumper wires in correspondence to the setup of the LABview program. Once the materials were gathered and prepared, the experiment, which was divided into three parts, was initiated.

The first part of the experiment involved taking data from the strain gauge when it was lying flat on the lab table surface. This part provided a necessary baseline for the data that was later used to determine proper sample windows. The strain gauge was connected to the proper ports on the Wheatstone bridge which was connected to the DAQ that was linked to a computer with a USBC chord. Once the strain gauge was lying flat on the table, the sample windows for ADC channels 0,1, and 2 were set to ±0.64V on the LABview program. The program was ran and the trim potentiometer was adjusted until the $V_{amp}$ reading in the program reached a value equal to or less than 0.009 volts [2]. Once the proper Vamp reading was achieved, a data point was saved and the program was terminated. The data was uploaded to a file with a .xls extension to be used for later analysis. This process was repeated for the remaining sample windows mentioned in the introduction.

The second part of the lab was to estimate the radius of the stool using the strain and feeler gauge. For this to be done, appropriate sample windows had to be chosen for each channel. For channel one, corresponding to *Vs* (voltage from B+ on WSB to power on DAQ), the data taken when the strain gauge was lying flat was analyzed to determine the smallest sample window where saturation did not occur. If the value for the mean of *Vs* matched the voltage of the sample window and the standard deviation for *Vs* was equal to zero, then saturation had occurred. The equations used by LABview to calculate the mean and standard deviation of voltages are shown below (6) and (7).

$$Mean\ V_x = \frac{\Sigma V_x}{N} \quad (6)$$

$$\sigma = \lim_{N\to\infty}\left(\frac{1}{N}\Sigma(x_i - \mu)^2\right)^{1/2} \quad (7)$$

For the mean equation, $\Sigma V_x$ is the sum of all voltages and *N* is the number of readings, and for the standard deviation equation, $\sigma$ is the standard deviation, $x_i$ is the individual sample and $\mu$ is the mean of all samples [4]. The smallest sample window without saturation was found to be ±2.56 volts. This made sense because the mean of *Vs* was about 2.5 ± 2.92 x $10^{-4}$ volts which fell just within that sample window. Next, the sample windows for channels two and three were determined using the equation below (8), where $\varepsilon$ is the strain, $\Delta V_g$ is the tared WSB voltage, $G_f$ is the gauge factor equal to 2.085 ± 0.0104, $V_s$ is the excitation voltage of the bridge equal to 2.5 ± 2.92 x $10^{-4}$ volts, $\Delta V_{amp}$ is the tared $V_{amp}$, and *gain* is the difference between $V_{amp}$ and $V_g$ equal to 900 ± 4.50 [5].

$$\varepsilon = \frac{4\Delta V_g}{V_s G_f} = \frac{4\Delta V_{amp}}{gain V_s G_f} \quad (8)$$

Before using (8) presented above, strain was solved using (4) with *ri* equal to 240 millimeters, *d* equal to 0.04 ± 0.02 millimeters and *t* equal to 0.4 ± 0.015 millimeters. With all other variables now defined, $\Delta V_g$ and $\Delta V_{amp}$ were solved and the smallest sample window that encompassed their values were chosen. For channel one corresponding to $\Delta V_g$,, ±0.64V was chosen as the sample window and for channel two corresponding to $\Delta V_{amp}$, ±2.56V was chosen as the sample window.

After setting the correct sample windows for all three channels on the LABview dashboard, the program was run with the strain gauge lying flat on the table. While the strain gauge remained flat, the trim potentiometer was adjusted until the value of $V_{amp}$ was equal to or less than 0.009 volts. Then the values for *d* and *t* defined in the previous paragraph were inputted into the LABview controls and a data point was saved [2]. While the program was still running, $V_g$ and $V_{amp}$ were tared by copying and pasting the current value of the indicator into the appropriate control corresponding to each variable [2]. The equation below (9) illustrates the tare operation, where $\Delta V_{amp}$ is the change in amplified voltage, $V_{amp}$ is the amplified voltage, and $V_{amp,tare}$ is the tare value being subtracted [5].

$$\Delta V_{amp} = V_{amp} - V_{amp,tare} \quad (9)$$

Although only the tared amplified voltage equation is shown, the same exact equation was applied to the WSB voltage just by switching every instance of $V_{amp}$ with $V_g$ . Once these variables were tared, the feeler gauge with the strain gauge attached was flexed around the footrest of the lab stool and fastened to it with masking tape as shown in Fig. 1 below.

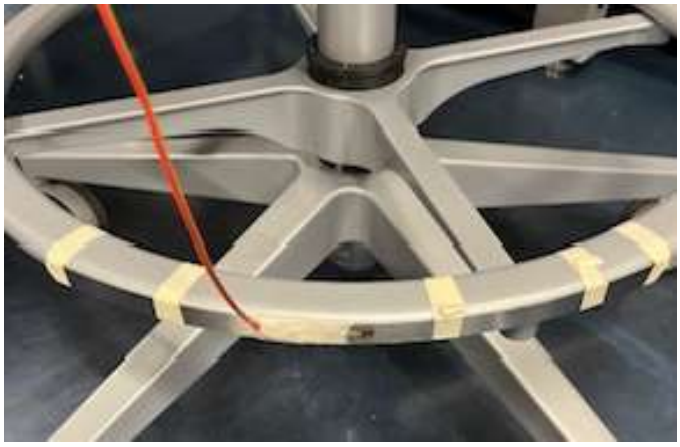

Fig. 1. Feeler gauge affixed to a strain gauge taped to the footrest of the lab stool.

Once the strain and feeler gauge were secured to the stool, multiple data points were collected, and the program was stopped. The data was saved to a file with a .xls extension.

The third and final part of the lab was to estimate the radius of the footrest of the lab stool manually. A flexible tape measure was used to find the circumference of the footrest with the aid of masking tape if necessary, and then (5) was used to approximate the radius [2]. Once all measurements were taken, any masking tape used for parts two and three of the lab were disposed of.

## III. Results

The resolution for each sample window was found using (1) and revealed that a smaller sample window corresponded to a decreased resolution (Table I).

Table I
Sample Window Resolution

| Sample Window (V) | Resolution |
|---|---|
| ±0.64 | $7.81 \times 10^{-5}$ |
| ±1.28 | $1.56 \times 10^{-4}$ |
| ±2.56 | $3.13 \times 10^{-4}$ |
| ±5.12 | $6.25 \times 10^{-4}$ |
| ±10.24 | $1.25 \times 10^{-3}$ |

In part one of the experiment, the LABview program calculated the mean (6) and standard deviation (7) of each of the three voltages while the strain gauge laid flat on the lab table. Data for $V_s$ (Table II), $V_g$ (Table III), and $V_{amp}$ (Table IV) were collected for each of the five sample windows.

Table II
Mean and Standard Deviation of $V_s$

| Sample Window (V) | Mean $V_s$ (V) | Standard Deviation $V_s$ |
|---|---|---|
| ±0.64 | 0.6399 ± 0.0000 | 0 |
| ±1.28 | 1.2798 ± 0.0000 | 0 |
| ±2.56 | 2.2114 ± $6.04 \times 10^{-4}$ | $3.08 \times 10^{-4}$ |

Table II Continued
Mean and Standard Deviation of $V_s$

| | | |
|---|---|---|
| ±5.12 | 2.2107 ± $6.78 \times 10^{-4}$ | $3.46 \times 10^{-4}$ |
| ±10.24 | 2.2090 ± $6.13 \times 10^{-4}$ | $3.13 \times 10^{-4}$ |

Table III
Mean and Standard Deviation of $V_g$

| Sample Window (V) | Mean $V_g$ (V) | Standard Deviation $V_g$ |
|---|---|---|
| ±0.64 | $7.21 \times 10^{-4}$ ± $1.14 \times 10^{-4}$ | $5.80 \times 10^{-5}$ |
| ±1.28 | $6.09 \times 10^{-4}$ ± $1.16 \times 10^{-4}$ | $5.90 \times 10^{-4}$ |
| ±2.56 | $3.22 \times 10^{-4}$ ± $1.02 \times 10^{-4}$ | $5.20 \times 10^{-5}$ |
| ±5.12 | 0.0012 ± $5.57 \times 10^{-4}$ | $2.84 \times 10^{-4}$ |
| ±10.24 | 0.0011 ± $8.98 \times 10^{-4}$ | $4.58 \times 10^{-4}$ |

Table IV
Mean and Standard Deviation of $V_{amp}$

| Sample Window (V) | Mean $V_{amp}$ (V) | Standard Deviation $V_{amp}$ |
|---|---|---|
| ±0.64 | 0.0027 ± 0.0073 | 0.0037 |
| ±1.28 | $2.00 \times 10^{-6}$ ± 0.0074 | 0.0038 |
| ±2.56 | $3.19 \times 10^{-4}$ ± 0.0075 | 0.0038 |
| ±5.12 | $4.10 \times 10^{-4}$ ± 0.0073 | 0.0037 |
| ±10.24 | 0.0012 ± 0.0074 | 0.0037 |

In part two of the experiment, the feeler gauge affixed to the strain gauge was bent around and taped to the footrest of the lab stool. The readings of $V_g$ (Table V) and $V_{amp}$ (Table 6) were tared while the gauge was still flat on the table and then new values were measured for these variables when the gauge was bent.

Table V
Tared and Bent Feeler Gauge $V_g$ Data

| Tared or Bent | Mean $V_g$ (V) | Standard Deviation $V_g$ |
|---|---|---|
| Tared (flat) | $2.35 \times 10^{-4}$ ± $1.00 \times 10^{-4}$ | $5.70 \times 10^{-5}$ |
| Bent Gauge | 0.0015 ± $1.00 \times 10^{-4}$ | $5.50 \times 10^{-5}$ |

Table VI
Tared and Bent Feeler Gauge $V_{amp}$ Data

| Tared or Bent | Mean $V_{amp}$ (V) | Standard Deviation $V_{amp}$ |
|---|---|---|
| Tared (flat) | 0.0039 ± 0.0073 | 0.0036 |

Table VI Continued
Tared and Bent Feeler Gauge $V_{amp}$ Data

| | | |
|---|---|---|
| Bent Gauge | 1.1282 ± 0.0075 | 0.0038 |

With the above voltage readings, the LABview program calculated strain and radius of the footrest (Table VII) using (8) and (4), respectively, for both $V_g$ and $V_{amp}$.

Table VII
$V_g$ and $V_{amp}$ Strain and Radius Data

| Voltage | Strain (mm/mm) | *ri* (mm) |
|---|---|---|
| $V_g$ | $9.52 \times 10^{-4} \pm 0.0012$ | 251.79 ± 39.00 |
| $V_{amp}$ | $9.59 \times 10^{-4} \pm 1.12 \times 10^{-5}$ | 250.06 ± 22.58 |

In part three of the experiment, the circumference of the footrest of the lab stool was measured manually with a tape measure and then radius was hand calculated using (5). The value of π was important to consider in this calculation as its uncertainty contributed to that of the radius, as shown in the results below (Table VIII).

Table VIII
Circumference and Radius from Tape Measure

| *C* (mm) | π | *ri* (mm) |
|---|---|---|
| $1.598 \times 10^{3} \pm 0.50$ | $3.1416 \pm 5.00 \times 10^{-5}$ | 254.33 ± 0.0796 |

## IV. Discussion

This experiment measured the radius, *ri*, of the footrest of a lab stool with three methods. These methods included using the voltage from the WSB, $V_g$, the amplified WSB voltage, $V_{amp}$, and a manual tape measure. For the first two methods, a strain gauge affixed to a feeler gauge was used to gather data. The proper sample windows for the voltage channels had to be chosen in the LABview program to ensure this data was not saturated. Each of the three methods yielded a different result and uncertainty.

The sample windows set the voltage range that the ADC within the DAQ could measure. Smaller sample windows provided a finer resolution (Table I), meaning each digital step represented a smaller voltage change, thus reducing rounding errors. Additional benefits of a fine resolution pertaining to this experiment include better sensitivity to subtle deformations in the feeler gauge and more accurate readings for small changes in voltage. However, the trade-off to these benefits was the risk of saturation occurring if the actual voltage value exceeded the range set by the sample window. Therefore, choosing the correct sample window was a balance between resolution and range, as shown in the analysis of $V_s$ below.

The excitation voltage of the Wheatstone bridge, $V_s$, was equal to 2.5 volts. Therefore, the reading for mean $V_s$ was saturated in sample windows ±0.64 and ±1.28 because these ranges did not contain that value. Therefore, the $V_s$ reading maxed out these sample windows with an uncertainty and standard deviation of zero because the value remained at the maximum for all samples (Table II). For the ±2.56 sample window and onwards, the mean $V_s$ reading represented its real value since it was within the window range, however, it was around 2.21 volts rather than 2.5 volts. The reason behind this error was because the sauter on a wire between the DAQ and WSB was in contact with another wire for a different channel. This caused a short that decreased the value of $V_s$. However, this value was still greater than 1.28 and less than 2.56, and therefore did not impact finding the correct sample window to choose for this voltage. The error with this sauter was corrected for part two of the lab and the value of $V_s$ remained equal to about 2.5 volts as intended. It is likely that the short caused additional inaccuracies in the flat data taken for $V_g$ (Table III) and $V_{amp}$ (Table IV), but fortunately this data did not impact the outcome of the experiment as correct sample windows were still selected based off hand calculations addressed in the procedure.

A comparison of the three radius results provides insight towards the accuracy and precision of the methods. The radius calculated using $V_g$ was 251.79 ± 39.00 millimeters, the radius calculated using $V_{amp}$ was 250.06 ± 22.58 millimeters, and the radius calculated from the tape measure method was 254.33 ± 0.0796 millimeters. All these values were overall similar in magnitude, suggesting all three methods were an appropriate means to measuring the radius of the footrest. However, the larger deviation came from the uncertainties. After applying the 95% confidence interval based off the uncertainties, the voltage methods had a very large range of values that overlapped, while the range of values for the tape measure method remained small.

The $V_g$ method presented the widest range of values suggesting it was the least precise method. When the voltage was amplified for the $V_{amp}$ method, the ADC became more sensitive to small variations and reduced the effects of slight fluctuations, leading to an overall smaller uncertainty and higher precision. However, compared to the tape measure method, the amplified voltage still lagged far behind in terms of producing precise results. Sources of the lower precision in the strain gauge methods compared to the tape measure method included a larger uncertainty stack up, sensor noise, improper taring, positioning errors, and flexing inconsistencies in the feeler gauge. While these sources came from both the devices and human error, the tape measure method only involved human error and was therefore more accurate and precise. Additionally, the tape measure method only involved a single, direct measurement which contributed to the minimal variation and uncertainty.

In the future, improvements could be made to the strain gauge methods to increase their precision. A more stable mounting method and refined taring technique could be used to reduce strain gauge noise. The strain gauge could be affixed to the feeler gauge in a better way than with glue and tape to increase the sensitivity of the strain gauge to feeler gauge movement. Finally, taking more data points could improve statistical accuracy and decrease the uncertainty in the strain gauge methods.

## V. Conclusion

The tape measure method was the most reliable estimate of the footrest radius as it yielded a result of 254.33 ± 0.0796 millimeters, while the $V_g$ and $V_{amp}$ methods resulted in radius measurements of 251.79 ± 39.00 millimeters and 250.06 ± 22.58 millimeters, respectively. Despite all three methods yielding a result of similar magnitude, the uncertainties of the strain gauge methods presented a much larger confidence interval range revealing they had a lower precision. This experiment emphasized the importance of concepts such as sample window selection, signal amplification and uncertainty analysis as all these topics impacted findings. Additionally, it highlighted the capabilities in data acquisition of devices such as DAQs, Wheatstone bridges and strain gauges. Further improvements such as reducing strain gauge sound and better mounting techniques could be applied to this experiment to increase the precision of results.

## Appendix

All parameters used in this experiment had an associated uncertainty. Some of the uncertainties were provided by manufacturers or known by rule of thumb (Table IX). Statistical uncertainties with a 95% confidence interval were found for all the voltage measurements recorded by the DAQ based off 1000 samples (Table X). The uncertainties for parameters dependent on other variables were found using RSS propagation equations that considered all contributing uncertainties (Table XI).

Table IX
Given Uncertainties

| Parameter (units) | Uncertainty | Source |
|---|---|---|
| $t$ (mm) | ±0.015 | Manufacturer |
| $d$ (mm) | ±0.02 | Manufacturer |
| $G_f$ (none) | ±0.0104 | Manufacturer |
| *gain* (none) | ±4.50 | Manufacturer |
| $C$ (mm) | ±0.50 | Rule of Thumb |

Table X
Statistical Uncertainties

| Parameter (units) | Uncertainty | Number of Samples |
|---|---|---|
| $V_s$ (V) | ±2.92 x $10^{-4}$ | 1000 |
| $V_g$ (V) | ±1.08 x $10^{-4}$ | 1000 |
| $V_{amp}$ (V) | ±0.0075 | 1000 |
| $V_{g,tare}$ (V) | ±1.14 x $10^{-4}$ | 1000 |
| $V_{amp,tare}$ (V) | ±0.0073 | 1000 |

Table XI
RSS Propagated Uncertainties

| Parameter | Uncertainty | Units |
|---|---|---|
| $\Delta V_g$ | ±1.60 x $10^{-4}$ | V |
| $\Delta V_{amp}$ | ±0.0105 | V |
| $ri\ V_g$ | ±39.00 | mm |
| $ri\ V_{amp}$ | ±22.58 | mm |
| $ri\ (C)$ | ±0.0796 | mm |

RSS Propagation Uncertainty Equations:

WSB Voltage Change ($\Delta V_g$)

$$\mu_{\Delta V_g} = [\frac{\partial \Delta V_g}{\partial V_g}^2 \mu_{V_g}^2 + \frac{\partial \Delta V_g}{\partial V_{g,tare}}^2 \mu_{V_{g,tare}}^2]^{1/2} \tag{10}$$

Amplified Voltage Change ($\Delta V_{amp}$)

$$\mu_{\Delta V_{amp}} = [\frac{\partial \Delta V_{amp}}{\partial V_{amp}}^2 \mu_{V_{amp}}^2 + \frac{\partial \Delta V_{amp}}{\partial V_{amp,tare}}^2 \mu_{V_{amp,tare}}^2]^{1/2} \tag{11}$$

WSB Voltage Strain

$$\mu_{\varepsilon_g} = [\frac{\partial \varepsilon_g}{\partial \Delta V_g}^2 \mu_{\Delta V_g}^2 + \frac{\partial \varepsilon_g}{\partial V_s}^2 \mu_{V_s}^2 + \frac{\partial \varepsilon_g}{\partial G_f}^2 \mu_{G_f}^2]^{1/2} \tag{12}$$

Amplified Voltage Strain

$$\mu_{\varepsilon_{amp}} = [\frac{\partial \varepsilon_{amp}}{\partial \Delta V_{amp}}^2 \mu_{\Delta V_{amp}}^2 + \frac{\partial \varepsilon_g}{\partial V_s}^2 \mu_{V_s}^2 + \frac{\partial \varepsilon_g}{\partial G_f}^2 \mu_{G_f}^2 + \frac{\partial \varepsilon_g}{\partial gain}^2 \mu_{gain}^2]^{1/2} \tag{13}$$

Radius WSB Voltage ($ri\ V_g$)

$$\mu_{ri_{V_g}} = [\frac{\partial ri_{V_g}}{\partial t}^2 \mu_t^2 + \frac{\partial ri_{V_g}}{\partial d}^2 \mu_d^2 + \frac{\partial ri_{V_g}}{\partial \Delta V_g}^2 \mu_{\Delta V_g}^2 + \frac{\partial ri_{V_g}}{\partial V_s}^2 \mu_{V_s}^2 + \frac{\partial ri_{V_g}}{\partial G_f}^2 \mu_{G_f}^2]^{1/2} \tag{14}$$

Radius Amplified Voltage ($ri\ V_{amp}$)

$$\mu_{ri_{V_{amp}}} = [\frac{\partial ri_{V_{amp}}}{\partial t}^2 \mu_t^2 + \frac{\partial ri_{V_{amp}}}{\partial d}^2 \mu_d^2 + \frac{\partial ri_{V_{amp}}}{\partial \Delta V_g}^2 \mu_{\Delta V_g}^2 + \frac{\partial ri_{V_{amp}}}{\partial V_s}^2 \mu_{V_s}^2 + \frac{\partial ri_{V_{amp}}}{\partial G_f}^2 \mu_{G_f}^2 + \frac{\partial ri_{V_{amp}}}{\partial gain}^2 \mu_{gain}^2]^{1/2} \tag{15}$$

Radius Circumference

$$\mu_{ri} = \frac{\partial ri^2}{\partial C} \mu_C^2 \tag{16}$$

Although not shown in the above equation, the uncertainty of the radius calculated from the circumference was also impacted by the uncertainty of pi as shown in the results.

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